

Inheritance and Method Overriding

1. Create a base class **Shape** :

- `__init__(self, name)`

- `area(self)` method (return 0 by default)

- `info(self)` method that returns name only

Create a subclass Rectangle that inherits from Shape:

Add attributes: length and width

Override `area()` to calculate and return the rectangle area (length × width)

Override `info()` to display name, dimensions, and area

2. Create a parent class **Employee**:

- `__init__(self, name, salary)`

- `calc_bonus(self)` method calc bonus of 10% of salary and return it

`display(self)` to display employee name and salary.

Create Two subclass that inherits from Employee. SoftwareEngineer and DevopsDeveloper.

Add Attributes - designation

Override `calc_bonus` method in two child class and calc bonus with 20% of salary for SoftwareEngineer and 15% of salary for DevopsDeveloper and return the bonus information along with employee information.

3. Create two base classes:

- Flyable** with `fly()` method

- Swimmable** with `swim()` method

Create class AmphibiousVehicle that inherits from both Flyable and Swimmable

Add name and speed attributes

Override both `fly()` and `swim()` methods with specific behavior

Add `perform_actions()` method that calls both overridden methods